

Quiz 8

On a recent homework assignment, you thought about a bag holding slips of paper with the numbers 1 through 4. Imagine the chance experiment of drawing two slips from the bag (without replacement). Let X be the higher of the two numbers drawn and Y be the lower of the two numbers drawn. If the two numbers drawn are equal, then X and Y are equal.

1. Make a table showing the joint distribution of X and Y .

$Y =$	$X = 1$	2	3	4	marginal for Y
1	0	$2/12$	$2/12$	$2/12$	$6/12$
2	0	0	$2/12$	$2/12$	$4/12$
3	0	0	0	$2/12$	$2/12$
4	0	0	0	0	0
marginal for X	0	$2/12$	$4/12$	$6/12$	

2. Add the marginal distributions for X and Y to your table.

3. Are X and Y independent random variables? Explain your answer using your table.

These random variables are not independent. For example, we have $P(X = 2) = 2/12$ and $P(Y = 2) = 4/12$, but $P((X = 2) \cap (Y = 2)) = 0 \neq (2/12)(4/12)$.

4. Calculate $E[X | Y = 1]$.

We calculated the expected value of X using the conditional probabilities $P(X = k | Y = 1)$. These probabilities are $P(X = 1 | Y = 1) = 0$, $P(X = 2 | Y = 1) = 1/3$, $P(X = 3 | Y = 1) = 1/3$, $P(X = 4 | Y = 1) = 1/3$. We thus find $E[X | Y = 1] = (1)(0) + 2(1/3) + 3(1/3) + 4(1/3) = 9/3 = 3$.